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# DP:-

① i/p  $\rightarrow$  arr  $\rightarrow$  { 1, 7, 10, 13, 14, 19 }

7, 13, 19

{ 1, 7, 13, 19 }

{ 1, 10 }

{ 1, 13 }

{ 1, 7, 13, 19 }

{ 1, 14 } { 1, 19 }

A.P

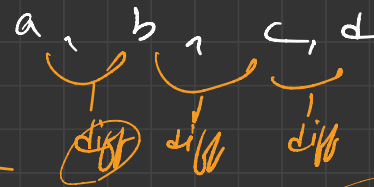
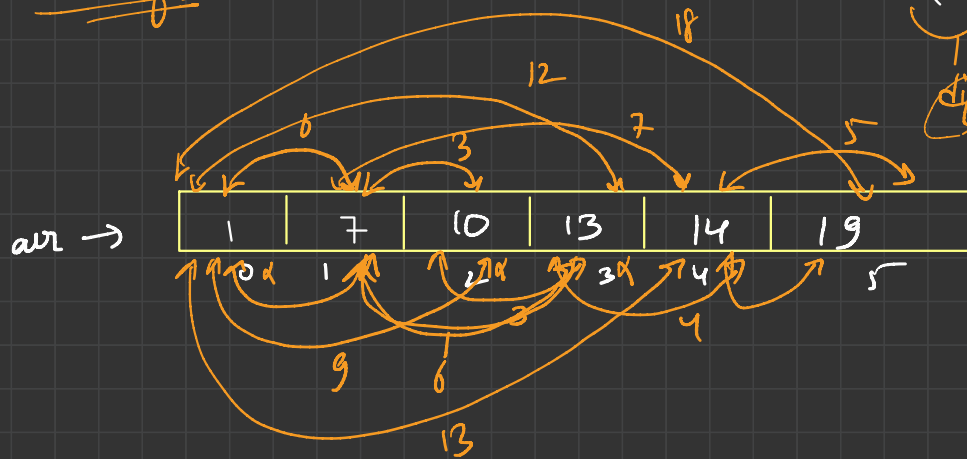
a  $\rightarrow$  first term  
d  $\rightarrow$  common diff.

a, a+d, a+2d, a+3d, a+4d, ... a<sub>n</sub>

i/p  $\rightarrow$  { 2, 4, 6, 8, 10 }  $\rightarrow$  Length = 5

2 2 2 2

# Brute force:

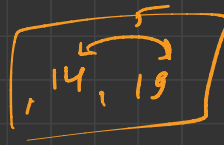
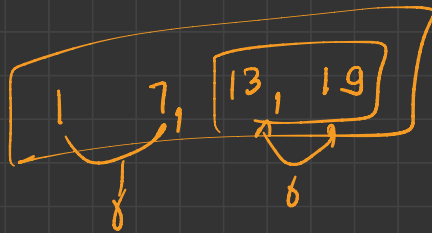


diff = 3

$n^2$  for ( $i=0$   
for ( $j=i+1$ ))

A.P. → sarri A.P.

→ 2 elements → A.P. → last element



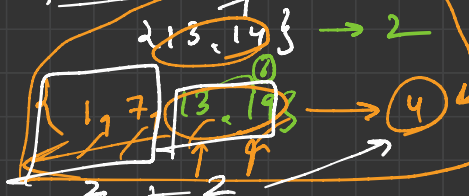
pass 3: - (3)

$$\{7, 10, 13\} \rightarrow 3$$

$$\{10, 14\} \rightarrow 2$$

$$\{1, 10, 19\} \rightarrow 3$$

pass 4: -



pass 5

$$\{14, 19\} \rightarrow 2$$

|   |   |    |    |    |    |
|---|---|----|----|----|----|
| 1 | 7 | 10 | 13 | 14 | 19 |
|---|---|----|----|----|----|

pass 1

$$\{1, 7\} \rightarrow 2$$

$$\{1, 10\} \rightarrow 2$$

$$\{1, 13\} \rightarrow 2$$

$$\{1, 14\} \rightarrow 2$$

$$\{1, 19\} \rightarrow 2$$

pass 2

$$\{7, 10\} \rightarrow 2$$

$$\{1, 7, 13\} \rightarrow 3$$

$$\{7, 14\} \rightarrow 2$$

$$\{7, 19\} \rightarrow 2$$

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Mem.

dp

fun  $\rightarrow$  }

dp [stor & return]

B.C  $\nrightarrow$  bad  $\rightarrow$  dp

$\downarrow$   
index

$\downarrow$   
diff

o/p

$\downarrow$

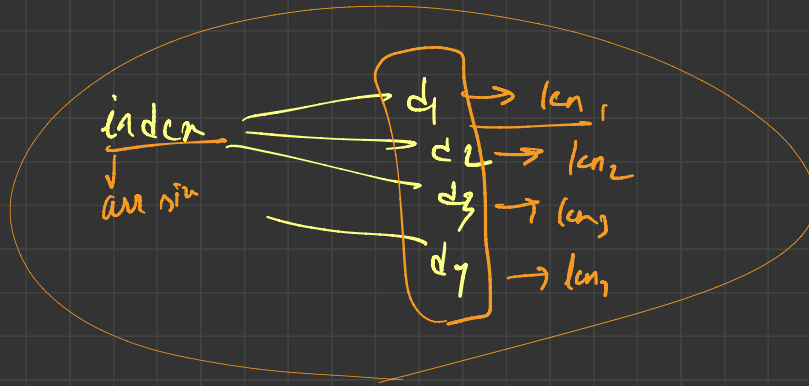
begin

dp [i] [diff]

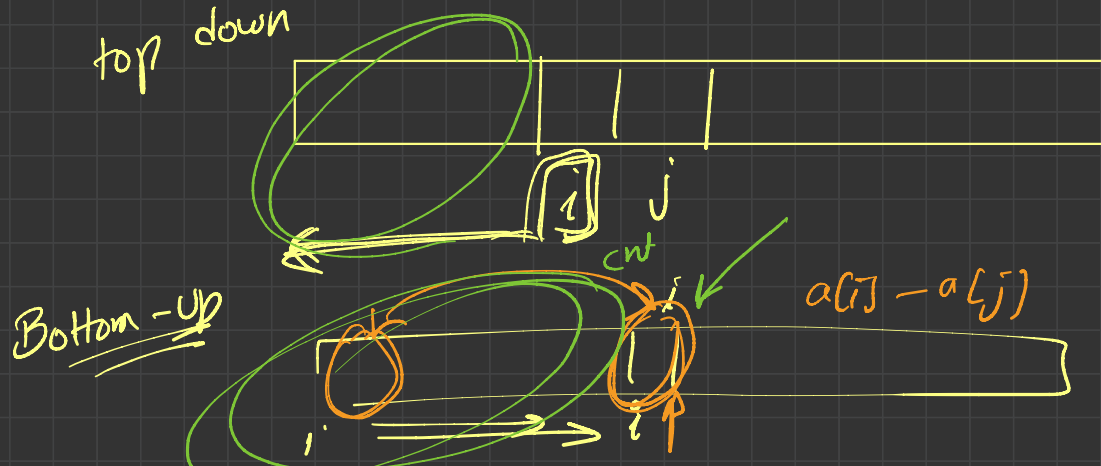
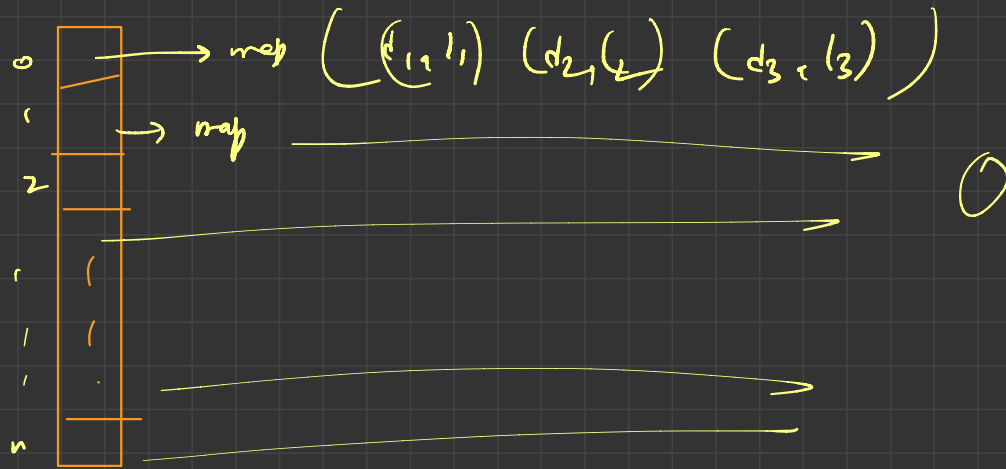
dp [i] [diff]

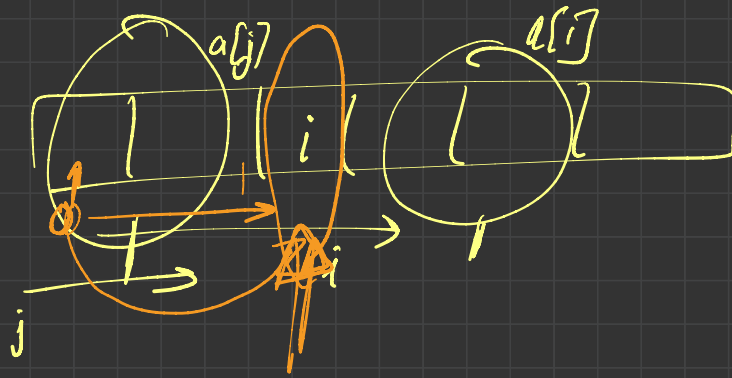
$index < arr\ size$

$dp [len + 1]$



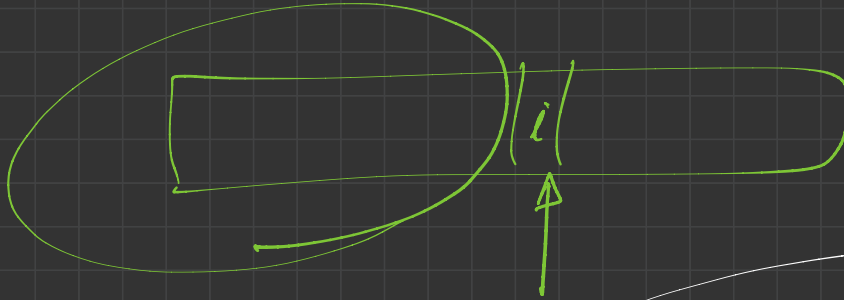
$unordered\_map <int, int> dp [len + 1]$





$$T.C \rightarrow n^2$$

$$S.C \rightarrow n^2$$



$$O(n)$$

$$a, b, c$$

$$\frac{a+c}{2} = b$$

$$a+c=2b$$

$$i \text{ --- } j$$

$$3 \text{ element AP}$$

$a, b, c$

$a, a+d, a+2d$

$$\frac{a + (a + 2d)}{a + c} = 2a + 2d = 2(a + d)$$

$$\frac{a + c}{2} = 2b$$























